TESTING AND EVALUATION OF SPACE GRADE SPACE-BORNE VIDEO CAMERA

LABORATORY FOR ELECTRO-OPTICS SYSTEMS

(LEOS-ISRO)

Training Report

Submitted in partial fulfilment of the requirements for the award of degree of

Bachelor of Technology

(Aerospace Engineering)

Submitted to

RV College of Engineering

From 15/11/2023 to 15/12/2023

SUBMITTED BY

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SUBMITTED TO

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(Professor & Head of Department)

Signature of the student

Signature of the HOD

CERTIFICATE

This is to certify that this project entitled **"TESTING AND EVALUATION OF SPACE GRADE SPACE-BORNE VIDEO CAMERA"** carried out by **Sonika Jain CS**, USN **1RV22AS054**, A bonafied student of **RV College of Engineering, Bengaluru, Karnataka,** towards partial fulfilment for the award of the degree of **Bachelor of Technology in Aerospace Engineering**, is a bonafied record of project work carried out by him/her from 15th November 2023 to 15th December 2023 under our supervision. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the Project report deposited in the school library. The Project report has been approved as it satisfies the academic requirements of the said Degree.

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STUDENT DECLARATION

To whom so ever it may concern

I**, Sonika Jain CS , 1RV22AS054** hereby declare that the work done by me on "**TESTING AND EVALUATION OF SPACE GRADE SPACE-BORNE VIDEO CAMERA**" from **November, 2023** to **December, 2023**, under the supervision of **Shri A. K. Sharma**, **Division Head MSOD, Laboratory For Electro Optics Systems(LEOS), ISRO,** and **Dr. Ravindra S Kulkarni**, **Professor & Head of Department,** Department of Aerospace Engineering, RV College of Engineering, Bengaluru, Karnataka, is a record of original work for the partial fulfilment of the requirements for the award of the degree, **Bachelor Of Technology (Aerospace Engineering)**.

Place: Bengaluru

Date: 15th December, 2023

Name: Sonika Jain CS

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Department of Aerospace Engineering

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DECLARATION BY THE SUPERVISORS

To whom so ever it may concern

This is to certify that **Sonika Jain CS, 1RV22AS054** from RV College of Engineering, Bengaluru, Karnataka, has worked as a trainee in **Laboratory For Electro Optics Systems(LEOS), ISRO** on "**TESTING AND EVALUATION OF SPACE GRADE SPACE-BORNE VIDEO CAMERA** " under my supervision from **November, 2023** to **December, 2023**. It is further stated that the work carried out by the student is a record of original work to the best of my knowledge for the partial fulfilment of the requirements for the award of the degree, **Bachelor of Technology (Aerospace Engineering)**.

A K Sharma

Head of MSOD,
LEOS – ISRO

Dr. Ravindra s Kulkarni

(Professor & Head of Department)

Signature of External Guide

Signature of Internal Guide

Dated:

Dated:

ACKNOWLEDGMENT

I take this opportunity to acknowledge all those who have helped me in the completion of my internship.

Firstly, I express my sincere gratitude to the **Laboratory For Electro-Optics Systems (LEOS)**, **ISRO** for giving me the opportunity to carry out the internship for five months at their premises.

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Lastly, I thank my college, **RVCE, my parents, and my friends** for their support and motivation always.

Executive Summary

The Thesis is organized as follows:

Part - 1

Chapter 1: Provides info of company's profile.

Chapter 2: Explains briefly about various camera and their applications.

Chapter 3: Provides design software description.

Chapter 4: Provides objectives, methodology and construction of Space Borne Camera.

Part - 2

Chapter 1: Provides information about setup and testing of the camera. This includes fixtures designed for testing

Chapter 2: Provides information about space environments tests implemented in order to simulate the working conditions in outer space.

Chapter 3: Brief note on Finite Element Analysis

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PART 1

CHAPTER 1:

1. INTRODUCTION OF THE COMPANY

1.1. INDIAN SPACE RESEARCH ORGANISATION (ISRO)

"There are some who question the relevance of space activities in a developing nation to us; there is no ambiguity of purpose. We do not have the fantasy of competing with the economically advanced nations in the exploration of the moon or the planets or manned space-flight. But we are convinced that if we are to play a meaningful role nationally, and in the community of nations, we must be second to none in the application of advanced technologies to the real problems of man and society."

- ***Dr. Vikram Sarabhai***

1.1.1. INTRODUCTION

The Indian Space Program was formally organized in 1972 when the government of India set up the space program with a view to promote the development and application of space technologies and space sciences for identified national objectives. The Indian Space Program is directed towards the goal of self-reliant use of space technology for national development. The main thrust are as follows:

- Satellite remote sensing for resource survey.
- Environmental monitoring and meteorological services.
- Development and operation of indigenous satellite and launch vehicle, providing their space services.

The main objective of ISRO is to develop space technology and its application to various national tasks. Accordingly, Indian Space Research Organization (ISRO) has successfully operationalized two major satellite systems namely Indian National Satellites (INSAT) for communication services and Indian Remote Sensing (IRS) satellites for management of natural resources; also, Polar Satellite Launch Vehicle (PSLV) for launching IRS type of satellites and Geostationary Satellite Launch Vehicle (GSLV) for launching INSAT type of satellites.

The Space Commission formulates the policies and oversees the implementation of the Indian space programme to promote the development and application of space science and technology for the socio-economic benefit of the country. DOS implements these programmes through, mainly Indian Space Research Organization (ISRO), Physical Research Laboratory (PRL), National Atmospheric Research Laboratory (NARL),

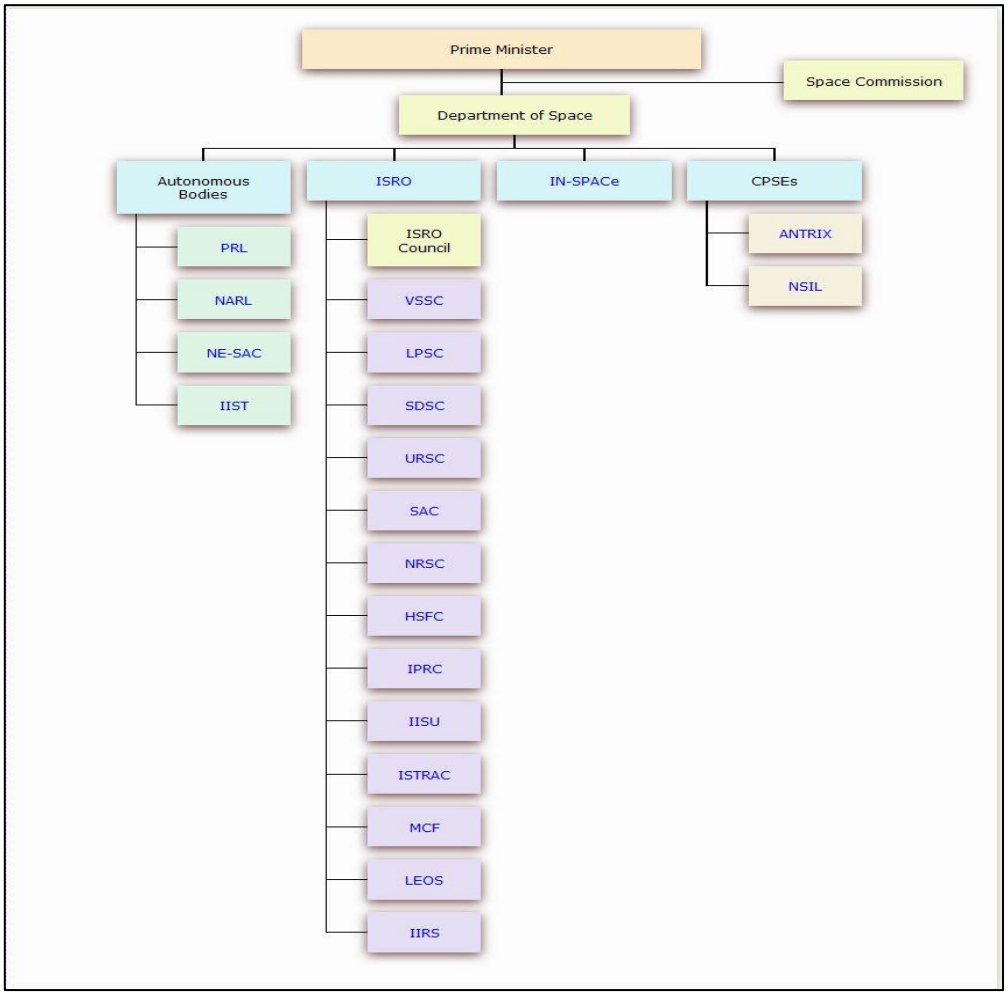


Figure 1: Flow chart of Department of space

North Eastern-Space Applications Centre (NE-SAC) and Semi-Conductor Laboratory (SCL).

1.2. ISRO CENTRES

- Vikram Sarabhai Space Centre **(VSCC)**: ISOR's single largest facility, near Trivandrum providing the technology base for launcher & propulsion development.

- Liquid Propulsion Systems Centre **(LPSC)**: Development branches in Bengaluru and Trivandrum are supported by major test facilities at Mahendragiri for wide spectrum of liquid motors, from reaction control system thrusters to the 720KN Vikas and Cryogenics Engines.

- URSC formerly known as ISRO Satellite Centre **(ISAC)**: ISRO's lead centre for the design, fabrication and testing science, technology, and application satellites.

- ISRO Inertial Systems Unit **(IISU)**: Provides inertial systems and components for satellites and launchers.

- **SHAR Centre**: ISRO's orbital launch site and largest solid motor production and test facility.

- ISRO Telemetry, Tracking & Command Network **(ISTRAC)**: Headquartered in Bengaluru, ISTRAC operates a network of ground station to provide TTC support for launcher and satellite operations.

- Space Applications Centre **(SAC)**: Located AT Ahmedabad, SAC is ISRO's applications R&D Centre, including communications, remote sensing and geodesy.

- Master Control Facility **(MCF)** at Hassan, 180 Km from Bengaluru.

1.3. LEOS - LABORATORY FOR ELECTRO-OPTICS SYSTEMS

Laboratory for Electro-Optics Systems is a research lab belonging to Indian Space Research Organisation. It involves in design and development of optics and sensor modules that can be deployed either on board satellite or with the launch vehicle.

Established in 1993, Laboratory for Electro-Optics Systems was established at the same place where the first Indian satellite Aryabhatta was fabricated in 1975, namely Bangalore. The laboratory has developed sensors for tracking Earth and Stars for the satellites which were launched when the space research was ushering in India. Satellites like Aryabatta, Bhaskara, Apple, IRS, SROSS and INSAT-2 have been equipped with the sensors developed by this laboratory. [1] The lab has also participated in the India's first moon mission Chandrayaan-1.

Figure 2: LEOS Office in Bengaluru

The Laboratory for Electro-Optics Systems (LEOS), Bengaluru is responsible for design, development and production of electro-optic sensors and optics for spacecraft use. Sensor system includes earth sensors, star sensors, sun sensors, magnetic sensors, fiber optic gyro, temperature sensors and processing electronics. Optics system includes optics for remote sensing camera, radiometers, star sensor optics, optical filter, optical masks, optical coatings, IR detectors and MEMS based inclinometer.

LEOS is actively involved in the development of new technologies for present / future satellites. This includes development active pixel sensor star tracker, Charge Coupled Device (CCD) based star tracker, Fibre Optics Gyro, Optical inter satellite link, high resolution camera optics, optical coatings and MEMS devices (magnetometer, accelerometer etc.). LEOS is situated at Peenya Industrial Estate, Bengaluru where the first Indian Satellite Aryabhata was fabricated in 1975.

1.4. APPLICATIONS

1.4.1. Earth Observation

With a humble beginning in early 60s, Indian space program has matured as a symbol of the country's sophisticated technological capabilities and its growing regional and global prestige. Over the last four decades, Indian Space program has made remarkable progress towards building the space infrastructure as the community resource to accelerate various developmental processes and harness the benefits of space applications for socio-economic development.

The Indian Space programme has the primary objective of developing space technology and application programmes to meet the developmental needs of the country. Towards meeting this objective, two major operational systems have been established – the Indian National Satellite (INSAT) for telecommunication, television broadcasting, and meteorological services and the Indian Remote Sensing Satellite (IRS) for monitoring and management of natural resources and Disaster Management Support.

1.4.2. Satellite Communication

The communication satellite series, which started with the APPLE satellite, grew into a very large constellation of satellites in the INSAT and GSAT series. These satellites revolutionized the technological and economic growth of the country. The INSAT satellite system is one of the largest domestic communication satellite systems providing regular services in the areas of telecommunications, business & personal communication, broadcasting, and weather forecasting & meteorological services. Today, newer initiatives have been taken to expand the INSAT applications to newer areas like Tele-education, Tele-medicine, Village Resource Centre (VRC), Disaster Management Support (DMS) etc., have enabled the space technology to reach the common man in India. The INSAT system has extended the outreach to less accessible areas like North- East, other far-flung areas and islands.

1.4.3. Disaster Management Support

Disaster management support, in terms of space based critical infrastructure and services, is yet another community centric deliverable. One of the elements on which the space-based Disaster Management Support (DMS) systems have been built is emergency communications systems. The DMS programme of ISRO/DOS, a convergence of space communications and remote sensing capabilities, is an effort to have technologically robust and a compatible system, which could strengthen India's resolves towards disaster management.

1.4.4. Satellite Navigation

Satellite Navigation service is an emerging satellite-based system with commercial and strategic applications. ISRO is committed to provide the satellite-based Navigation services to meet the emerging demands of the Civil Aviation requirements and to meet the user requirements of the positioning, navigation and timing based on the independent satellite navigation system. To meet the Civil Aviation requirements, ISRO is working jointly with Airport Authority of India (AAI) in establishing the GPS Aided Geo Augmented Navigation (GAGAN) system. To meet the user requirements of the positioning, navigation and timing services based on the indigenous system, ISRO is establishing a regional satellite navigation system called Indian Regional Navigation Satellite System (IRNSS).

1.4.5. Climate and Environment

ISRO has designed and developed indigenous systems for ground-based observations of weather parameters. It includes (i) Automatic Weather Station (AWS) to providing hourly information on critical weather parameters such as pressure, temperature, humidity, rainfall, wind and radiation from remote and inaccessible areas, (ii) Agro Metrological (AGROMET) Towers to measure soil temperature, soil moisture, soil heat and net radiation, wind speed, wind direction, pressure and humidity, (iii) Flux Tower for multi-level micrometeorological observation as well as subsurface observations on soil temperature and moisture over the vegetative surfaces, (iv) Doppler Weather Radar (DWR) to monitor severe weather events such as cyclone and heavy rainfall.

1.5. LAUNCHERS

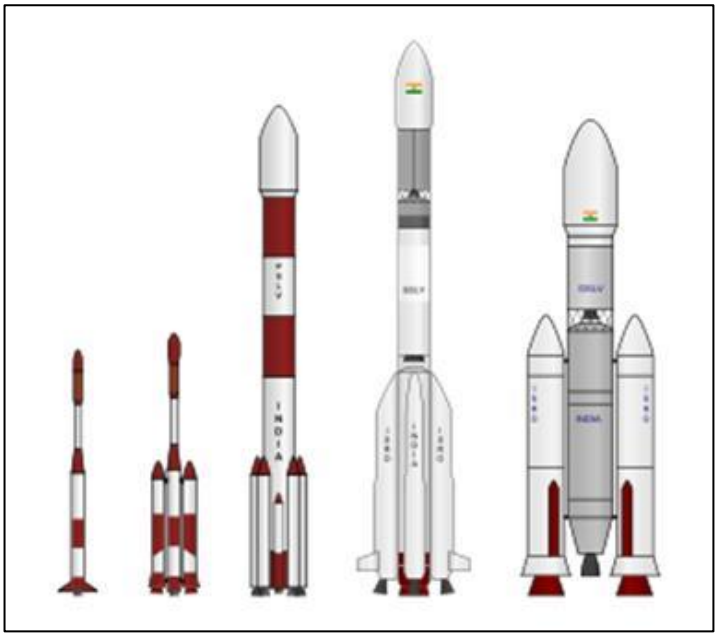


Figure 3: *Indian carrier rockets.*

Left to right: SLV, ASLV, PSLV, GSLV, GSLV Mark III

During 1960s and 1970s, India initiated its own launch vehicle programme owing to geopolitical and economic considerations. In the 1960s–1970s, the country successfully developed a sounding rockets programme, and by the 1980s, research had yielded the Satellite Launch Vehicle-3 and the more advanced Augmented Satellite Launch Vehicle (ASLV), complete with operational supporting infrastructure. ISRO further applied its energies to the advancement of launch vehicle technology resulting in the creation of PSLV and GSLV technologies.

1.5.1. Satellite Launch Vehicle (SLV)

The Satellite Launch Vehicle, usually known by its abbreviation SLV or SLV-3 was a 4-stage solid-propellant light launcher. It was intended to reach a height of 500 km and carry a payload of 40 kg. Its first launch took place in 1979 with 2 more in each subsequent year, and the final launch in 1983. Only two of its four test flights were successful.

1.5.2. Augmented Satellite Launch Vehicle (ASLV)

The Augmented Satellite Launch Vehicle, usually known by its abbreviation ASLV was a 5-stage solid propellant rocket with the capability of placing a 150 kg satellite into Low Earth Orbit. This project was started by the ISRO during the early 1980s to develop technologies needed for a payload to be placed into a geostationary orbit. Its design was based on Satellite Launch Vehicle. The first launch test was held in 1987, and after that 3 others followed in 1988, 1992 and 1994, out of which only 2 were successful, before it was decommissioned.

1.6. Currently Active

1.6.1. Polar Satellite Launch Vehicle (PSLV)

The Polar Satellite Launch Vehicle, commonly known by its abbreviation PSLV, is an expendable launch system developed by ISRO to allow India to launch its Indian Remote Sensing (IRS) satellites into Sun synchronous orbits. PSLV can also launch small satellites into geostationary transfer orbit (GTO). The reliability and versatility of the PSLV is proven by the fact that it has launched, as of 2014, 71 satellites/spacecraft (31 Indian and 40 foreign) into a variety of orbits. The maximum number of satellites launched by the PSLV in a single launch is 10, in the PSLV-C9 launch on 28 April 2008 (690 kg CARTOSAT-2A, 83 kg Indian Mini Satellite, and 8 Nano-satellites, launched by PSLV's "core-alone" version).

1.6.2. Geosynchronous Satellite Launch Vehicle (GSLV)

The Geosynchronous Satellite Launch Vehicle, usually known by its abbreviation GSLV, is an expendable launch system developed to enable India to launch its INSAT-type satellites into geostationary orbit and to make India less dependent on foreign rockets. At present, it is ISRO's second-heaviest satellite launch vehicle and is capable of putting a total payload of up to 5 tons to Low Earth Orbit. The vehicle is built by India with the cryogenic engine purchased from Russia while the ISRO develops its own engine programme.

In a setback for ISRO, the attempt to launch the GSLV, GSLV-F07 carrying GSAT-5P, failed on 25 December 2010. The initial evaluation implies that loss of control for the strap-on boosters caused the rocket to veer from its intended flight path, forcing a programmed detonation. Sixty-four seconds into the first stage of flight, the rocket began to break up due to

the acute angle of attack. The body housing the 3rd stage, the cryogenic stage, incurred structural damage, forcing the range safety team to initiate a programmed detonation of the rocket.

1.6.3. Geosynchronous Satellite Launch Vehicle Mark-III (GSLV III)

GSLV-Mk III can launch four tonne satellite into geosynchronous transfer orbit. It is a three-stage vehicle with a 110-tonne core liquid propellant stage (L-110) and a strap-on stage with two solid propellant motors, each with 200 tonne propellant (S-200). The upper stage will be cryogenic with a propellant loading of 25 tonne (C-25). It has a lift-off mass of about 640 tonnes and is 43.43 meters tall. The payload fairing has a diameter of 5 meters and a payload volume of 100 cubic meters. It will allow India to become less dependent on foreign rockets for heavy lifting.

CHAPTER 2:

2. VARIOUS SATELITE PACKAGES

2.1 VARIOUS SPACE BORNE CAMERA AND THEIR APPLICATIONS

These are just a few examples of the categories of camera used in space and their applications. Each type of camera serves a specific purpose and contributes to our understanding of the universe, exploration of celestial bodies, and advancement of scientific knowledge.

2.1.1 Imaging Camera:

These camera are used to capture high-resolution images of celestial bodies, spacecraft, and other objects in space. They are essential for scientific research, documentation, and public outreach.

2.1.2 Video Camera:

Video camera are used to record live footage of space missions, spacewalks, and other significant events. They provide valuable visual data for analysis, training, and public engagement.

2.1.3 Scientific Camera:

These specialized camera are designed to capture specific wavelengths of light or other forms of radiation. They are used for scientific experiments, such as studying the composition of celestial bodies or measuring atmospheric conditions.

2.1.4 Navigation Camera:

Navigation camera are used to determine the position and orientation of spacecraft by capturing images of stars, planets, or other reference points. They help in spacecraft navigation and ensure accurate trajectory calculations.

2.1.5 Telescopic Camera:

Telescopic camera are equipped with powerful lenses or mirrors to capture detailed images of distant objects in space. They are used for astronomical observations, studying galaxies, stars, and other celestial phenomena.

2.1.6 Stereo Camera:

Stereo camera use two or more lenses to capture images from slightly different angles. These images can be combined to create 3D models of objects or terrain on celestial bodies, aiding in mapping and exploration.

2.1.7 Time-lapse Camera:

Time-lapse camera capture a series of images at regular intervals, allowing for the observation of long-term changes in space, such as the movement of planets, the rotation of celestial bodies, or the growth of phenomena like auroras.

2.2 ACCELEROMETER

An accelerometer is a device that measures the vibration, or acceleration of motion, of a structure. The force caused by vibration or a change in motion (acceleration) causes the mass to “squeeze” the piezoelectric material which produces an electrical charge that is proportional to the force exerted upon it. Since the charge is proportional to the force, and the mass is constant, then the charge is also proportional to the acceleration. These sensors are used in a variety of ways – from space stations to handheld devices – and there’s a good chance you already own a device with an accelerometer in it. An accelerometer works by utilizing an electromechanical sensor that is designed to measure either static or dynamic acceleration.

Static acceleration is the constant force acting on a body, like gravity or friction. These forces are predictable and uniform to a large extent. For example, the acceleration due to gravity is constant at 9.8 m/s – and the gravitation force is almost the same at every point on Earth.

Dynamic acceleration forces are non-uniform, and the best example is vibration or shock. A car crash is an excellent example of dynamic acceleration. Here, the acceleration change is sudden when compared to its previous state. The theory behind accelerometers is that they can detect acceleration and convert it into measurable quantities

2.3 LASER ALTIMETER

Laser altimeter sensors are a class of devices, developed for airborne and space-based applications, with a primary role of measuring the topography of the Earth and other planetary surfaces. Laser altimeters are mounted for observation on aircraft or spacecraft platforms and it provides a high-resolution, high-accuracy measurement of the elevation and horizontal variability of planetary surfaces. The basis of the altimeter measurement is the timing of the round-trip propagation of short-duration pulses of laser radiation between the platform and the surface. Vertical (i.e. surface elevation) resolution of the altimetry measurement is determined primarily by laser pulse width, surface-induced spreading in time of the reflected pulse, and the timing precision of the altimeter electronics. Laser divergence angle and altimeter platform height above the surface determine the laser footprint size at the surface. On the other hand, laser pulse repetition-rate, laser transmitter beam configuration, and altimeter platform velocity determine the spacing between successive laser pulses. Typical sensor footprint diameters for an aircraft altimeter application are 1-10 m and are ~ 100 m for a spacecraft altimeter. Ideally, the pulse-rate is adjusted to make a contiguous profile of topography so that horizontal resolution is equal to the sensor footprint size. Successful operation of the laser altimeter sensor requires additional information on the laser altimeter platform pointing attitude and position 1. Recent major advances in laser gyros, star cameras, and kinematic positioning with the Global Positioning System permit recovery of platform effects to a sufficient level (e.g. 5-500 μ rad for pointing angle and 10 cm - 1 m for position) to enable sub-meter laser altimetry for the operating altitudes and velocities of both aircraft and spacecraft platforms.

CHAPTER 3:

3. DESIGN SOFTWARE DESCRIPTION

Designing mainly involves Part Modelling and Assembly of given components. It encompasses part modelling from 2D drawings given from industry and assembling those parts. The main objective is to generate 3D model of the assembly that should fulfil the given requirements. It concentrates mainly on Finite Element Method for validation of design and provides conclusion about the Space worthiness of the design. The main objective of the project is to prepare FE model of the Package by performing geometry clean-up, mesh generation, add boundary conditions and loads, and simulate the model to get the results.

Fundamental modelling functions of automatic and manual mesh generation, application of loads and boundary conditions and model development and checking can be done using *NX Advanced FEM*. A robust set of visualization tools generates displays quickly, lets you view multiple results simultaneously and enables you to easily print the display. In addition, extensive post-processing functions enable review and export of analysis results to spreadsheets and provide extensive graphing tools for gaining an understanding of results. Some of its features are:

3.1 Core Modelling:

The core modelling capabilities of NX continue to improve in key areas, including feature templates, algorithmic modelling, and sketch. New features such as thread propagation across multiple bodies and advanced shape design through Draw Shape are powerful enhancements to your team's productivity and efficiency.

3.2 NX for BIM: This new add-on gives your designers access to the realm of Building Information Modelling (BIM) in the same NX environment as your existing workflows. Quickly create and document building with full 3D parametric design capabilities.

3.3 NX Assemblies: You'll find multiple productivity and efficiency gains for NX Assemblies in this release, such as Simplify Assembly, Partition in Session, and Worksheets capabilities.

3.4 Model-Based Definition (MBD): The model-based definition capabilities of NX continue to help you build the most comprehensive digital twin. Areas of interest for this release include component PMI, automated PMI, feature template support, and technical data package (TDP).

3.5 Simulation Driven Design: NX Design Space Explorer includes integrations with Simcenter for closed-loop workflows. Topology optimization now provides for vibration analysis and analysis constraints.

3.6 NX Composites: Manage end-to-end composite design and development with the full power of an NX application. Simulate ply draping, rapidly iterate darting strategies, interrogate composite stackup, and export DXF files for nesting and cutting.

To realise the objective of the assigned project work i.e. to generate Part models and assembly model from the given 2D drawings and perform Finite Element Modelling to validate the design of the Package. The following methodology is adopted through the use of various features of NX.

- Geometrical modelling of components in NX 2212.
- Drafting of the modelled parts.
- Selection of materials.
- Assembly of modelled parts for FEA.

CHAPTER 4:

4. SPACE BORNE CAMERA

Space borne camera is basically a Color Camera onboard to capture various space activities. These cameras are used to record live footage of space missions, spacewalks, and other significant events. They provide valuable visual data for analysis, training, and public engagement. Here in this project, it is used for monitoring events of docking, undocking of modules, antennae deployment from 25 meter up to 0.5 meter. Several logical schemes have been worked out which compasses provision to control crucial parameters like exposure time, resolution, gain, Fps. Their collective ability involves capturing high-quality images in the challenging conditions of space.

Figure 4 Space Borne Camera

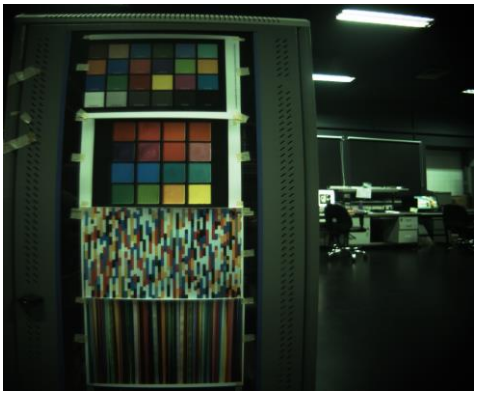


Figure 5 Color chart image taken by space borne camera

Construction of Space Borne Camera:

The significance of these components lies in the detector which converts light into electrical signals, optics focus and direct the light, electronics control the camera's functions, and the mechanical structure ensures stability and protection. All these elements must be carefully designed, tested, and integrated to create a reliable and effective space-borne camera for capturing invaluable scientific data and images from the cosmos.

4.1 Detector:

The detector, typically a CCD (Charge-Coupled Device) or CMOS (Complementary Metal-Oxide-Semiconductor) sensor, is the heart of the camera. It converts incoming light into electrical signals, which are then processed to create an image. The quality and sensitivity of the detector are crucial for capturing clear and detailed images in space

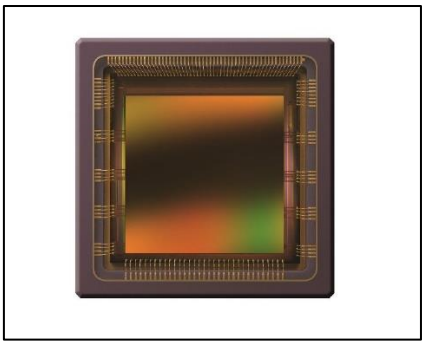


Figure 6: CMOS COLOUR DETECTOR

Specifications of Detector:

Parameters	Specifications	Remarks
Detector	CMOS Bayer Colour Detector	CMV4000
No. of Pixel	512x512	Full sensor is 2048x2048. To maintain the video frame rate 512x512 pixels are used.

4.2 Optics:

Optics refers to the lenses and other optical components that focus and direct the incoming light onto the detector. The design and quality of the optics play a critical role in determining the resolution, field of view, and image quality of the camera. In space, where precise focusing is essential, optics must be carefully designed and calibrated for optimal performance.

Figure 7 Edmund optics

Specifications of Optics:

Parameters	Specifications	Remarks
FOV@8mm C-Mount Optics	69°+/0.5°@ Full Circular	2/3 Inch Optical Format,

4.2.1 OPTICAL ABERRATIONS

Space-borne camera can encounter several optical aberrations due to the challenging conditions and constraints of space environments. Some common optical aberrations include:

1. Chromatic Aberration: Chromatic aberration occurs when different wavelengths of light are focused at different distances by the lens, resulting in color fringing or blurring at the edges of objects.

2. Spherical Aberration: Spherical aberration occurs when light rays passing through the edges of a lens or mirror are focused at a different point compared to the rays passing through the center.

3. Coma: Coma is an aberration that causes off-axis points of light to appear comet-shaped or elongated.

4. Astigmatism: Astigmatism causes points of light to appear elongated or distorted due to the variation in focus between meridional and sagittal planes.

5. Field Curvature: Field Curvature refers to the curvature of the image plane or focal plane.

6. Distortion: Distortion causes straight lines to appear curved or bent in images.

4.3 Electronics:

The electronics in a space-borne camera are responsible for controlling and operating the various functions of the camera. They handle tasks such as adjusting exposure settings, digitizing the electrical signals from the detector, and transmitting the data back to Earth. The electronics must be radiation-hardened to withstand the harsh radiation environment in space and ensure reliable operation.

PCB Operations:

PCB performs the following functions like generating different clocks for various logics and driving the detector, store images and transmit image data.

4.4 Mechanical Structure:

The mechanical structure of a space-borne camera provides support, stability, and protection to the sensitive components. It includes the housing, mounting structures, and mechanisms for focusing, pointing, and stabilizing the camera. The mechanical structure must be lightweight yet robust enough to withstand the vibrations, shocks, and extreme temperatures encountered during launch, operation, and re-entry.

Description:

The mechanical structure of the camera is designed such that the optics is place at the right distance to the imaging plane and the right amount of space is accommodated to stack the PCBs.

Specification:

Operating Temperature: -35°C to +50°C

Material used: AA6061-T6

Part – II

CHAPTER 1

BASIC FUNCTIONALITY CHECK OF CAMERA

1.1 FOCUSING

Focusing tests in space-borne camera are conducted to ensure that the camera's optics are properly calibrated and adjusted for optimal image sharpness and clarity. These tests are essential because achieving accurate focus in space can be challenging due to factors such as microgravity, thermal variations, and limited access to manual adjustments.

1.1.1 SETUP

1. The 2-Axis turn table (Annexure -2) is mounted on a rigid surface.
2. On the turn table, fixture is attached upon which camera is mounted and torqued.
3. A chess board pattern is setup at a distance of 1 meter on a 1-Axis translator.

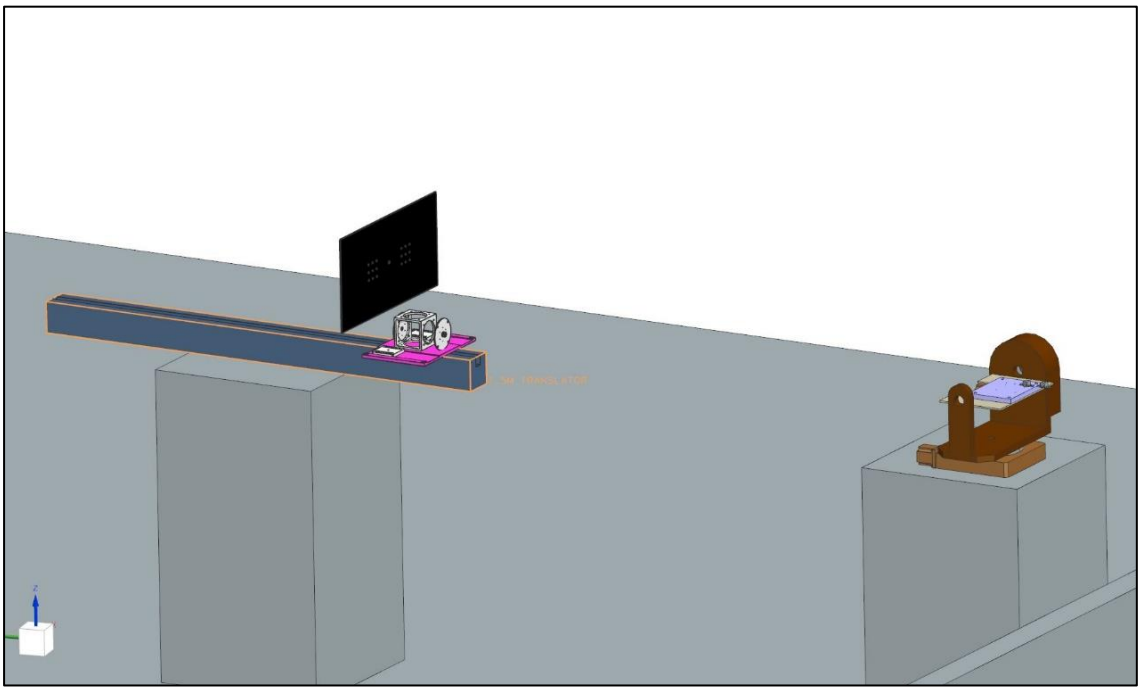


Figure 8: Setup for alignment of space borne camera

1.1.2 PROCEDURE

- A chess board pattern is placed in front of the camera, with a fixed distance from the lens of the camera to the chess board pattern.
- Then camera is turned ON and images of the chess board are captured.
- Images are then checked for clarity near the white and black squares of the chess board pattern.
- If image is defocused the lens of the camera is changed and images will be taken again till a well-focused image is found.
- When image looks focused the lens will be locked by potting the rings.

1.2 ALIGNMENT

Alignment in space-borne camera is crucial for achieving optimal optical performance, high-resolution imaging, accurate spatial measurements, well-defined field of view and perspective, reliable calibration, and structural stability. Through careful alignment procedures, space agencies and mission teams can ensure that the camera deliver high-quality, scientifically valuable images and data from the challenging environment of space.

Reasons:

- Optical Performance
- Resolution and Spatial Accuracy
- Field of View and Perspective
- Calibration and Calibration Stability
- Structural Integrity and Stability

1.2.1 SETUP

1. A light with a pinhole is placed in front of the camera at the same distance where it is focused.
2. In between an Auto-Collimator is placed, in order to align the light coming out of the pinhole to the lens of the camera
3. Position of the camera is fixed i.e., there will be no change in its height and translations positions.
4. Auto-Collimator will be aligned to the camera, by transferring the camera's reference to the auto-collimator.

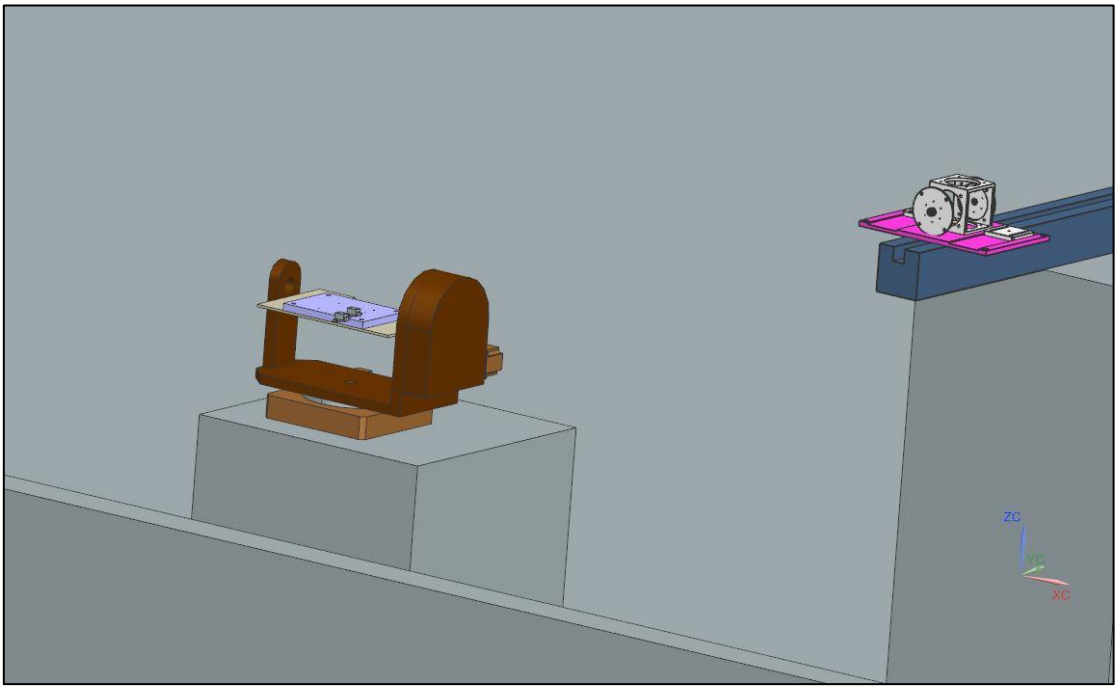


Figure 9: Setup for alignment of space borne camera

1.2.2 PROCEDURE

- Auto-collimator will be rotated 180 degrees in azimuth, facing the pinhole.
- Position of the auto-collimator will be fixed with the reference of the lens of the camera and the pinhole will be aligned to the auto-collimator by changing its height and translation.
- When Auto-collimator is looking into the pinhole, the position of pinhole is locked and verified with lens of camera.

1.3 FOV CHECK

FOV checks in space-borne camera are crucial for mission success, target acquisition, image planning, resource allocation, image registration, data interpretation, and instrument performance validation. By verifying the camera's field of view, mission teams can ensure that the camera captures the desired targets, optimizes resource utilization, and provides accurate and valuable data for scientific analysis and exploration in space.

1.3.1 SETUP

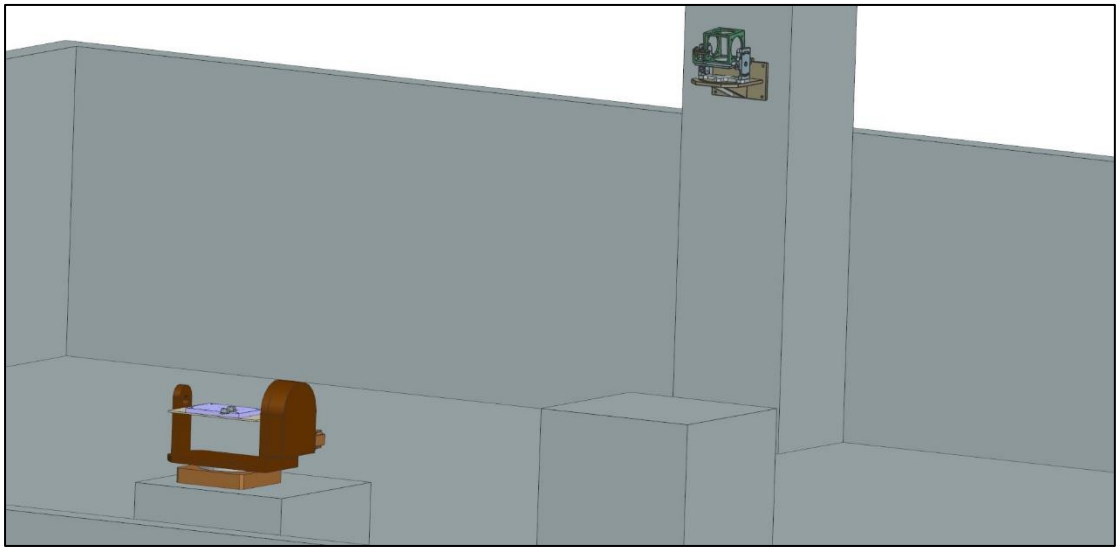


Figure 10 Setup for FOV check

1. The 2-Axis turn table (Annexure -2) is mounted on a rigid surface.
2. On the turn table, fixture is attached upon which camera is mounted and torqued.
3. A light source is fixed in front of the camera device under test and the camera which is mounted on the motion simulator is rotated for a wide angle along horizontal and vertical axes passing through the CG of the camera.

1.3.2 PROCEDURE

- A wall mountable two axis degree of freedom fixture was made, which supports azimuth and elevation changes.
- This fixture is to hold the pinhole light on to a wall which can also act as a fixed reference for alignment purposes.
- Camera will be facing the pinhole aligned with an auto-collimator as done in alignment.
- Azimuth and elevation position of the pinhole will be locked and not to be disturbed.
- Now that camera and pinhole light are aligned, the camera will be changing its azimuth in degrees.
- This will continue until camera cannot see the light coming out of pinhole.
- This will be done in all 4 directions of the camera establishing Field-Of-View of camera.

CHAPTER 2:

SPACE WORTHINESS TESTING FOR IMAGING CAMERA

2.1 VIBRATION

Vibration testing is an important step in the development and qualification process of space-borne camera. It is performed to assess the camera's ability to withstand the harsh mechanical vibrations experienced during launch and operation in space. Here's why vibration testing is done for space-borne camera:

2.1.1 MOTIVES

- **Launch Environment Simulation:**

The intense vibrations generated during rocket launch can potentially damage or affect the performance of sensitive equipment like space-borne camera. Vibration tests aim to simulate the launch environment by subjecting the camera to mechanical vibrations of varying frequencies and amplitudes. This helps ensure that the camera can withstand the vibrations encountered during launch without experiencing any adverse effects.

- **Structural Integrity Evaluation:**

Vibration testing allows engineers to evaluate the structural integrity and robustness of the camera. The high levels of vibration during launch can induce dynamic stresses on the camera's components, such as lenses, image sensors, and electronic circuitry. By subjecting the camera to controlled vibrations, any structural weaknesses or potential failure points can be identified and rectified before launch. This helps ensure the camera's reliability and longevity in space.

- **Image Quality Assessment:**

Vibration testing also helps assess the impact of mechanical vibrations on the image quality produced by the camera. Vibrations can introduce image blur, distortion, or misalignment of optical elements, which can degrade the overall image quality. By conducting vibration tests, engineers can evaluate the camera's ability to maintain image sharpness, stability, and optical performance under vibrational forces. This ensures that the camera can deliver high-quality images in the demanding space environment.

- **Electromechanical Performance Verification:**

Space-borne camera consist of various electromechanical components, such as motors, actuators, and mechanisms. Vibration testing helps verify the performance and functionality of these components under dynamic conditions. It ensures that the camera's mechanical systems, including focusing mechanisms, filter wheels, or image stabilization systems, can operate as intended despite the vibrations. This is crucial for achieving accurate image acquisition and functionality during space missions.

- **Failure Identification and Mitigation:**

Vibration testing allows engineers to identify potential failure modes and weaknesses in the camera system. By subjecting the camera to controlled vibrations, any unexpected behaviour, component failures, or loose connections can be detected. This enables engineers to make design improvements, implement necessary reinforcements, or apply additional damping measures to mitigate the identified risks. Ultimately, this helps enhance the camera's reliability and performance in space.

2.1.2 Types of Vibration tests

2.1.2.1. Sinusoidal Vibration

1. In sinusoidal vibration, the frequency range will increase gradually with time.

2. The frequency will sweep from 5Hz to 2000Hz.

3. Every component has different natural frequency, when the frequency of the fixture reaches the component's natural frequency, the component tends to resonate.

4. The peaks of the component and the fixture are compared.

2.1.2.2. Random Vibration

1. In Random vibration, all the frequencies from 5Hz to 2000Hz will occur at the same time.

3. The entire package reacts to their respective frequencies, providing us with an output which includes the response for every frequency subjected to.

4. The peaks of the component and the fixture are compared.

2.1.3 PROCEDURE

- Camera is mounted on a fixture which is specifically designed for vibration tests.
- The camera torqued before resuming the test.
- Vibration will be done in all 3 axes i.e., X, Y and Z.
- Accelerometers are placed at critical locations of the camera to get an understanding on how the camera reacts at every frequency.

CHAPTER 3:

Finite Element Analysis

3.1. INTRODUCTION TO FEA:

Finite Element Analysis, also known as Finite Element Method (FEM) was first developed to be used in aerospace and nuclear industries, where the safety of the structures (design of physical configuration of the craft to withstand the forces during the flight is critical. In recent years, there is tremendous growth in the usage of this method, which is directly attributable to the rapid advances in computer technology.

As a result, commercial finite element packages exist that can solve the most sophisticated problems, not only structural analysis, but in a wide range of phenomena those are: the steady state and dynamic temperature distributions, fluid flow and manufacturing processes such as injection moulding and metal forming.

FEA is the numerical simulation of the mechanical components acquired by discreating engineering components into finite number of building blocks (known as Elements) and by analysing those components for their acceptability and reliability.

3.1.1. Methods to solve any Engineering Problem are:

1. Analytical Method

2. Numerical Method

3. Experimental Method

3.1.2. Procedure for Solving any Analytical or Numerical Problems

There are 2 steps

Step 1) Writing governing equation - Problem Definition or in other words formulating the problem in the form of a mathematical equation

Step 2) Mathematical solution of governing equation

Result is summation of step 1 & step 2. Result will be 100 % accurate when there is no approximation at either of the steps (Analytical method).

Numerical methods make approximation at step 1 as well as at step 2 & hence al1 the numerical methods are approximate.

3.1.3. How does numerical methods solve the problem?

Numerical methods like FEM are based on discretization of integral form of equation. Basic theme of al1 numerical methods is to make calculations at only limited number of points & then interpolate the results for entire domain (surface or volume). Even before getting the solution we assume how the unknown is going to Vary over a domain. Say for example, when meshing is carried out using linear quadrilateral elements, assumption made is linear variation of displacement over the domain and for 8 nodded quadrilateral elements, assumption is parabolic variation. This may or may not be the case in real life & hence al1 numerical methods are based on an initial hypothetical assumption. After getting the results there are several ways to check numerical as well as practical or field result correlation accuracy & minimization of errors.

3.2. HISTORY

FEA was first developed in the year 1943 by **R. Couranit**, who utilized RITZ method of numerical analysis and minimization of variation calculus to obtain the approximate solutions to vibration systems. Shortly after the development, a paper published in the year 1956 by **Mr. M. J. Turner, R. W. Clough, H. C. Martin and L. J. Topp** established a broader definition of numerical analysis. FEA was limited due to expansion mainframe computers that are generally owned by the industries. FEA has been developed to an incredible precision. For instance, today's supercomputers can produce accurate results for all kinds of parameters.

3.3. ADVANTAGES of FEA

1. Visualization is increased
2. Design cycle time is decreased
3. Number of prototypes and Testing is decreased
4. Optimal design

3.4. STEPS OF FEA

- **Pre-processing**

In this stage the entire model of the structure is created and the material properties, loads and the constraints are specified.

- **Processing**

This involves computations of results such as displacements, stresses, etc. The data describing the FEM model is processed by the software, to produce the output results such as components displacements, accelerations, etc.

- **Post-processing**

This involves interpretations of results generated during the processing stage like deformed shape of the structure, graphical representation of stresses, and displacements. The animation can also be appointed during the process.

3.5 FINITE ELEMENT MODELLING

3.5.1. Introduction to FE Modelling

The analysis for stress and deformation of the loading of simple geometric structures can usually be accomplished by closed-form techniques. As the structures become more complex, the analyst is forced to use approximations of closed-form solutions, experiments, or numerical methods. There are many numerical techniques used in engineering applications for which digital computers are very useful. In the field of structural analysis, the numerical techniques generally employ a method which discretizes the continuum of structural system into a finite collection of nodes / elements. The technique used currently is the finite element method (FEM). In the finite element method of analysis, a complex region defining a continuum is discretized into simple geometric shapes called finite elements. The material properties and the governing relationships are considered over these elements and expressed in terms of unknown values at element corners. An assembly process duly considering the loading and constraints results in a set of equations. Solutions of these equations give the approximate behaviour of the continuum. Following are the methods employed to create finite element model of the Electronic Package.

3.5.2. Geometry Clean-up

Before meshing, certain features like fillets, chamfer, small holes can be removed. And, if the meshing is carried out manually, surface modification might be necessary in few regions. Figure 4.1 shows the geometry clean-up process. The fillet is removed for analysis purpose if and only if area is not critical. Also, it shows that a node is formed at the edge of the geometry ensuring good mesh.

Surface modification includes stitching a surface to another, filling the gaps, creating washer splits for critical holes, joining open/free edges, removing duplicate surfaces etc.,
Mesh quality increases and meshing can be done effortlessly if geometry clean-up is carried out properly before meshing.

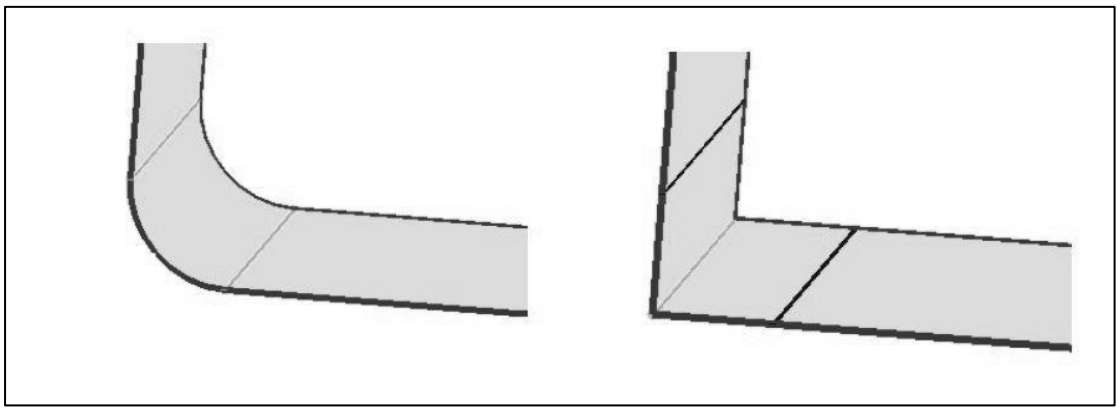


Figure 11: Removal of fillet for Analysis Purpose

3.5.3. Meshing

So, coming to the next topic, meshing. Here it raises so many questions like:

1. What is Meshing in FEA? So, the answer is: Meshing is one of the most important steps in performing an accurate simulation using FEA. A mesh is made up of elements which contain nodes (coordinate locations in space that can vary by element type) that represent the shape of the geometry. An FEA solver cannot easily work with irregular shapes, but it is much happier with common shapes like cubes. Meshing is the process of turning irregular shapes into more recognizable volumes called "elements."

2. Why do we carry out Meshing? So, the answer is: Basic theme of FEA is to make Calculations at only limited (Finite) number of points and then interpolate the results for entire domain (surface or volume). Any continuous object has infinite degrees of freedom and it's just not possible to solve the problem in this format. FEM reduces degrees of freedom from infinite to finite with the help meshing (Nodes and Elements).

3.6 RESULTS AND DISCUSSIONS

3.6.1 Modal Analysis

Modal analysis is technique which is used to examine the structure under dynamic excitation. The prime objective of this analysis was to find out the natural frequency and mode shapes of the Electronic Package structure for first 10 modes and to verify the stiffness and dynamic interaction between the components. Modal analysis solves an equation for which there is no applied load.

The FE Model which is shown in chapter 4 is simulated in NX-Nastran. Modal analysis mainly depends on three physical parameters of the component such as mass, stiffness and boundary conditions of structure. The structure is constrained at one end and free at the other end. So, this structure is now considered as cantilever beam. At constrained region, all degrees of freedom (Translational and Rotational) are fixed. Hence the first six modal frequencies in free modal analysis must be approximately equal to zero. After applying the boundary conditions, the FE model is checked for errors, if no errors found it is solved for modal frequencies. The solver calculates Eigen values i.e., modal frequencies.

3.6.2 Modal Analysis Results and Discussions

Modal analysis was carried out using the NX- Nastran for Electronic Package assembly; the results were observed and by analysing critical to quality characteristics

Free - Free Modal

Free-Free Modal analysis

In free-free modal analysis is done without application of boundary conditions. The Eigen frequencies in mode-1 to mode-6 is <0.0023 Hz for the assembly which confirms the accuracy of the FE model. The Mode-7 natural frequency is 377.97Hz for Modal assembly.

Here are some of the meshed and analysed ground components:

FEM OF CONNECTING JOINT

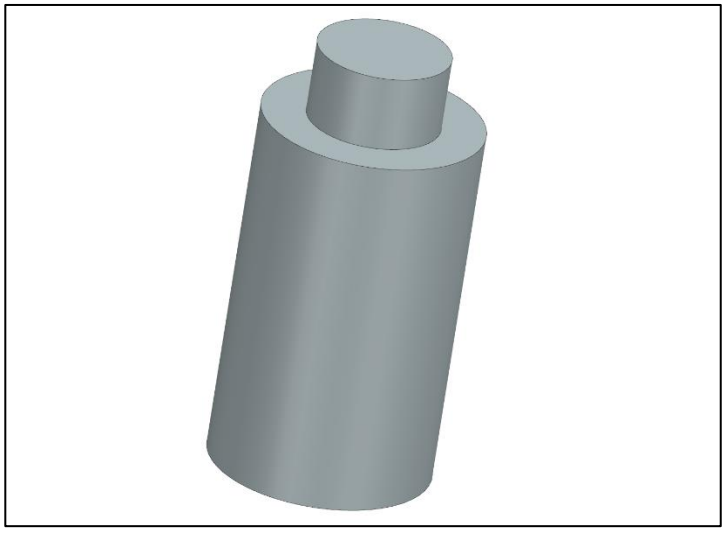


Figure 12 CAED Model of connecting joint

Dimensions (mm): Dia 5 X 10

Material: AA6061-T6

Mass: 0.5 gms

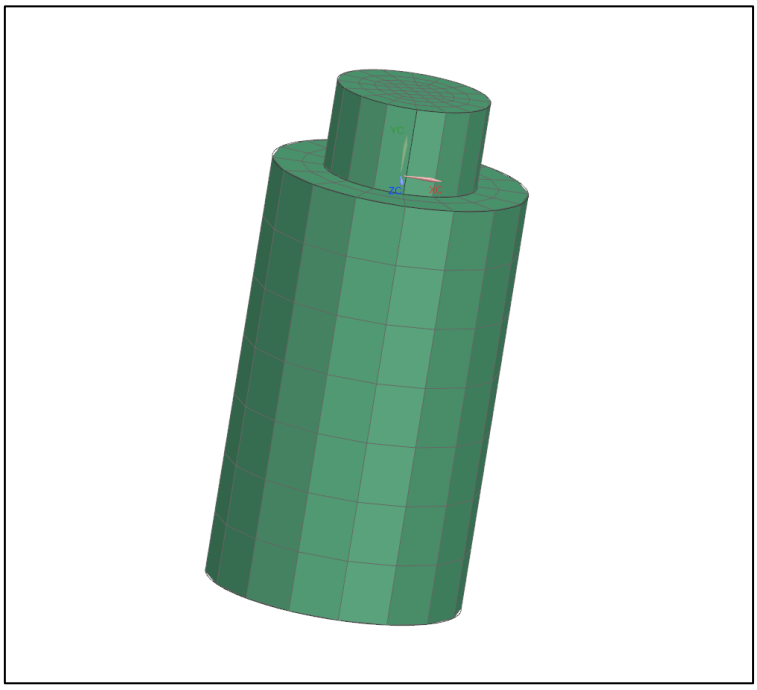


Figure 13: FE Model of connecting joint

Mesh Used: 3D Hex Mesh

Number of Elements: 736

Number of nodes: 913

Figure 14: ***Von-Mises Stress at natural frequency of the model***

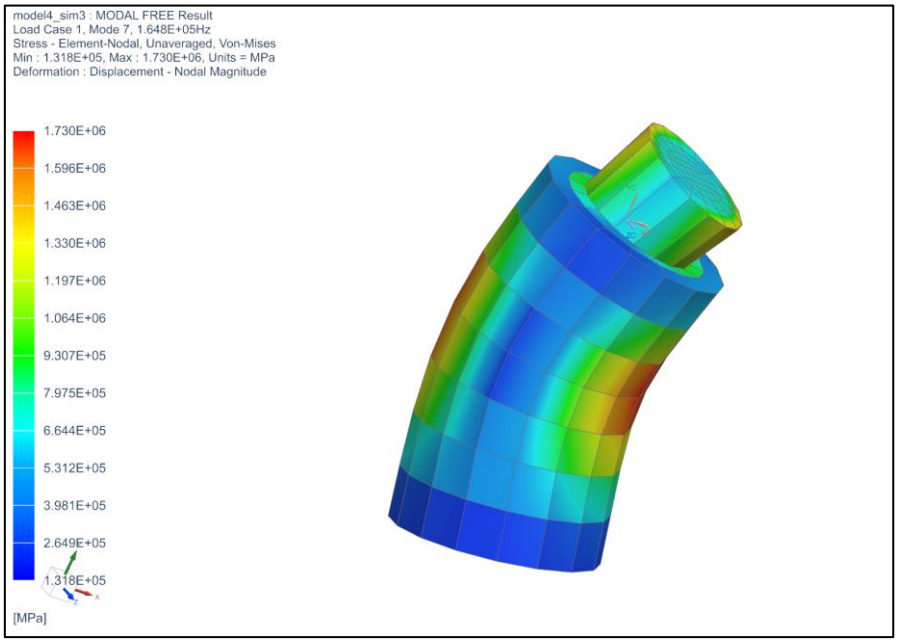
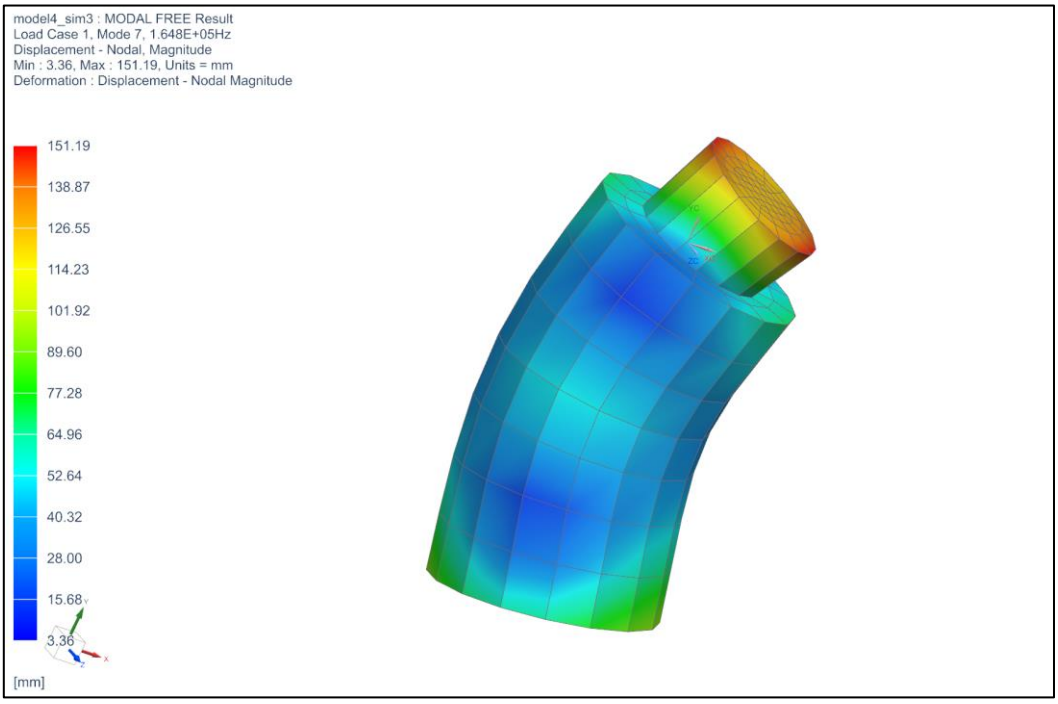


Figure 15: ***Displacement at natural frequency of the model***



Analysis Type: Free-Free

Natural Frequency: 1.6×10^5 Hz

FEM of 1 METER TITANIUM ROD

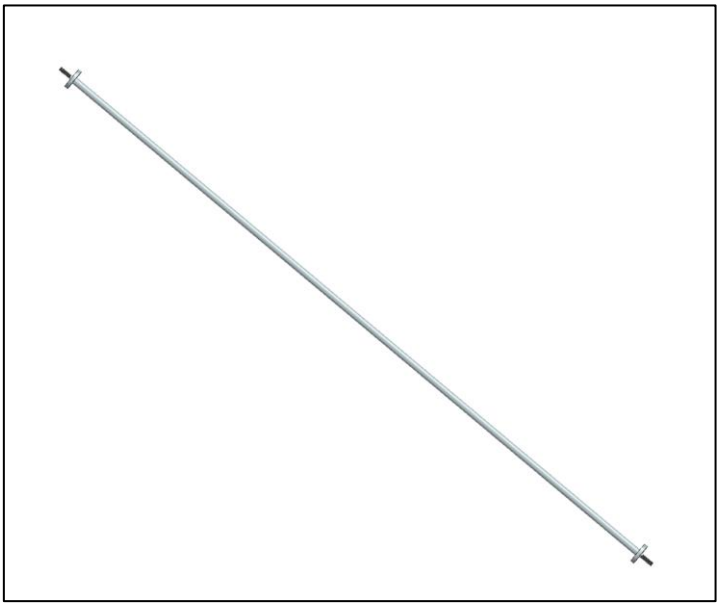


Figure 16 CAED Model of 1 meter titanium rod

Dimensions (mm): Dia 10 X 1040

Material: Ti4Al6V

Mass: 668 Gms

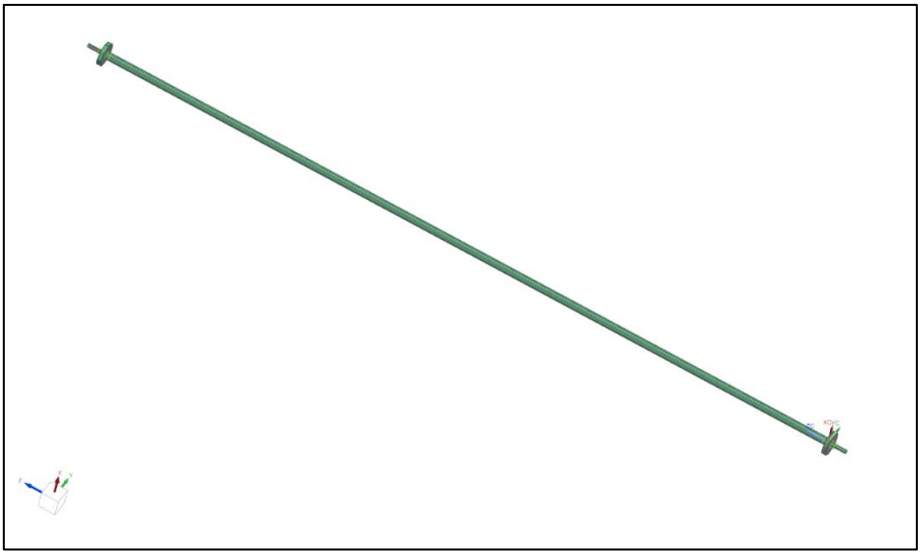


Figure 17: FE model of titanium rod

Mesh Used: 3D Hex Mesh

Number of Elements: 2732

Number of nodes: 4182

Figure 18: ***Von-Mises Stress at natural frequency of the model***

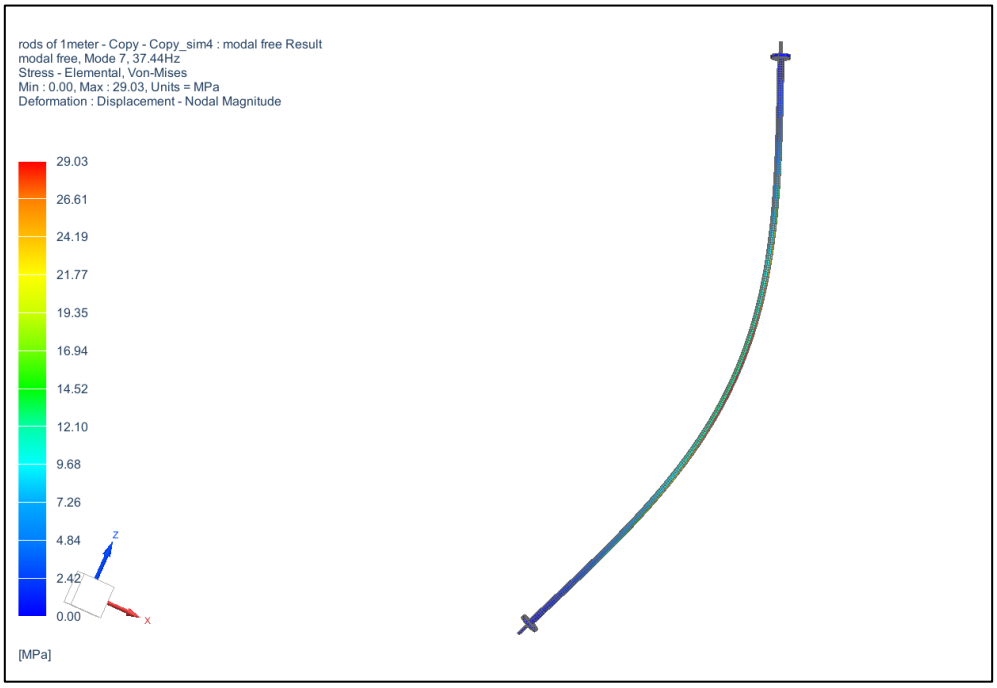
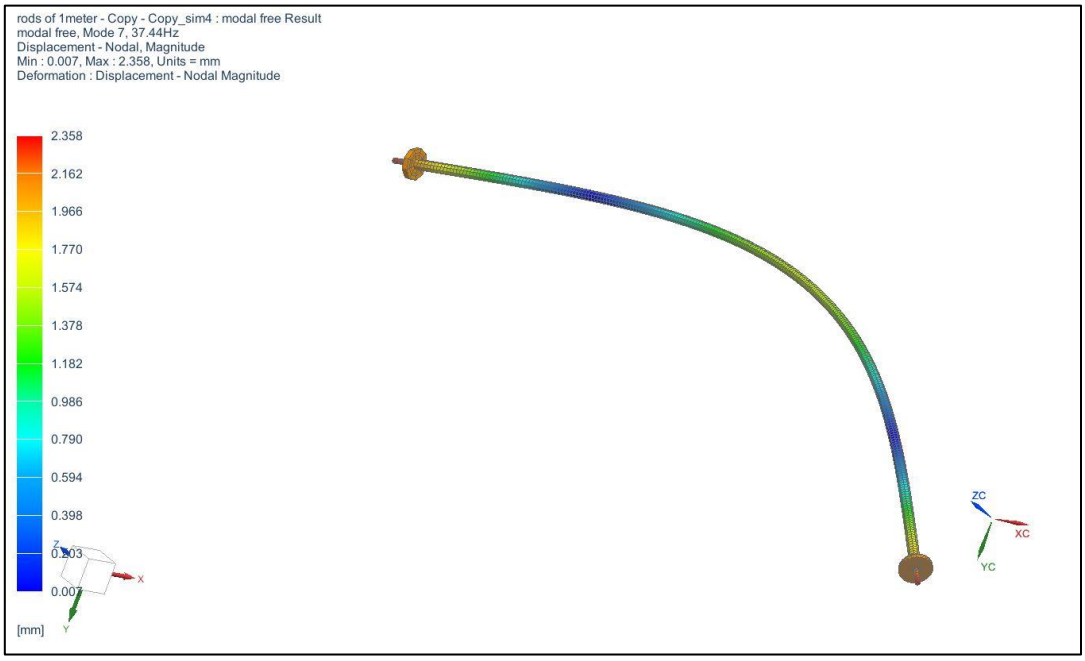


Figure 19: ***Displacement at natural frequency of the model***



Analysis Type: Free-Free

Natural Frequency: 37.44 Hz

3.7 FUTURE OF FEA

Research related to solution techniques & post processing tools is more or less saturated and in future focus will be mainly towards reducing the, improving quality & flow pattern of mesh generated by auto meshing. Also one can expect very powerful & user friendly commands for brick meshing (hopefully auto brick meshing too!).

Now in Indian companies are competent with structural analysis. In near future demand will before high-end analyses like NVH, CFD, Fatigue and Crash.
For reducing meshing time apart from conventional types of elements, research is also going on about polyhedral meshing and meshless (or mesh free) analysis. 2-d hexagonal elements divide a given surface into regions of equal area with least total perimeter.

While 3-d polyhedral elements are more efficient due to more faces shared by neighbouring elements (tetra has4while brick has 6, typically polyhedral has 10 and more faces). Advantage of polyhedral meshing could be described as less meshing the (supports automatic mesh like tetra elements) with higher accuracy (like hex elements) and that too at a smaller number of dofs. In short it combines advantage of tetra and hex elements. Some of the CFD software's have already started providing polyhedral meshing option. But now it would be too early to answer whether polyhedral mesh would be a regular feature of structural as well other types of analyses also.

Mesh free analysis is an attempt to integrate CAD and Solution steps directly i.e. without conventional & time-consuming linking step "meshing': There are several theories coming up based on creating calculation points during CAD geometry preparation itself, using integrals based on big blocks created automatically by software's and using a special method based on Finite Difference Method. One thing is for sure after 10 to 15 years, conventional meshing would be either very fast or easy or otherwise there will be no need of meshing at all.

CHAPTER 3:

CONCLUSION AND SCOPE FOR FUTURE WORK

3.1 CONCLUSIONS

Satellites employ a diverse array of sensors to gather essential data for various purposes, ranging from Earth observation to scientific research. Optical sensors, like multispectral and hyperspectral imagers, have proven crucial for monitoring vegetation, land cover changes, and climate patterns. Radar sensors, on the other hand, enable all-weather imaging and play a vital role in terrain mapping and disaster response.

Infrared sensors contribute significantly to atmospheric studies and temperature monitoring, aiding in weather forecasting. Furthermore, radio frequency sensors are instrumental in communication and navigation tasks, ensuring seamless satellite operations. The integration of these sensors allows satellites to provide comprehensive and accurate information for diverse applications.

As we conclude, it's evident that the continuous evolution of sensor technologies enhances the capabilities of satellites. The synergy between different sensors enables a holistic approach to data collection, contributing to advancements in climate science, agriculture, disaster management, and beyond.

3.2 SCOPE FOR FUTURE WORK

The future of satellite sensors holds exciting prospects with ongoing advancements in technology. Miniaturization and increased efficiency of sensors are anticipated, allowing for the deployment of smaller and more cost-effective satellites. Enhanced spectral resolution and sensitivity in optical sensors will provide finer details for Earth observation, aiding precision agriculture and environmental monitoring.

Artificial intelligence and machine learning algorithms will play a pivotal role in processing the vast datasets acquired by sensors, improving data interpretation and decision-making. Integration of advanced sensors, such as hyperspectral and LiDAR, will further refine 3D mapping and provide new insights into Earth's topography and vegetation structure.

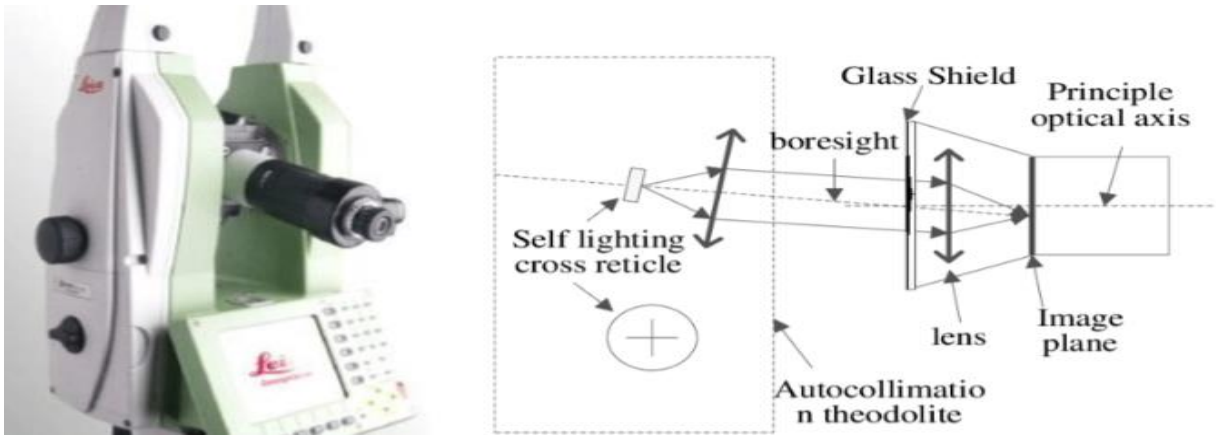
Additionally, advancements in quantum sensors hold promise for revolutionizing satellite-based measurements, offering unparalleled precision in various fields. As we look ahead, the collaboration between sensor technology, data analytics, and AI will unlock new frontiers in satellite applications, contributing to a more sustainable and informed understanding of our planet and beyond.

ANNEXURE – I

INSTRUMENTS USED

AUTOCOLLIMATOR

High-precision opto-mechanical structures have historically been used for both hardware and the associated alignment and assembly process, to maintain precision from tightly toleranced, machined flight hardware to reusable, modular tooling, especially in space applications. Theodolite auto collimation metrology is essential for spacecraft optical alignment, which uses optical instruments to place or determine the orientation and envelope of critical components on spaceflight hardware. An autocollimator is an optical instrument for collimating optical instruments and non-contact measurement of angles. They are typically used to align components and measure deflections in optical or mechanical systems



Theodolite Autocollimator

Figure 20: ***Autocollimator***

A theodolite is a small, movable optical alignment instrument that can be mounted on a tripod, with two perpendicular axes of rotation, one vertical and one horizontal. Motorized, sub-micron

precision actuators are used to adjust optical alignment in the autocollimator in two rotational degrees of freedom. The circles of rotation are precisely calibrated to mark the angle of rotation about each axis, thus providing the angular orientation of the ACM. Theodolite autocollimation measurements are used to determine the relative alignment between various components on a test object with respect to a common coordinate system (CS). Generally, the optical axis of each component would be previously related to an external optical reference surface, such as a mirror or an optical reference cube mounted rigidly to the component.

Autocollimation occurs when collimated light emanating from the theodolite is returned along the same path after its reflection from the reflective surface (mirror or cube face). A level of skill is required by theodolite operators to gain line-of-sight and to collimate on various reflective surfaces at various heights and angles that may be required to measure all required cube faces of components on a given test object.

2-AXIS TURN TABLE

The scope of the work covers Supply of 2-Axis Gimbal Mount (Accuracy - 1 arc second, Payload Capacity - 10 kg). Specifications of the item:

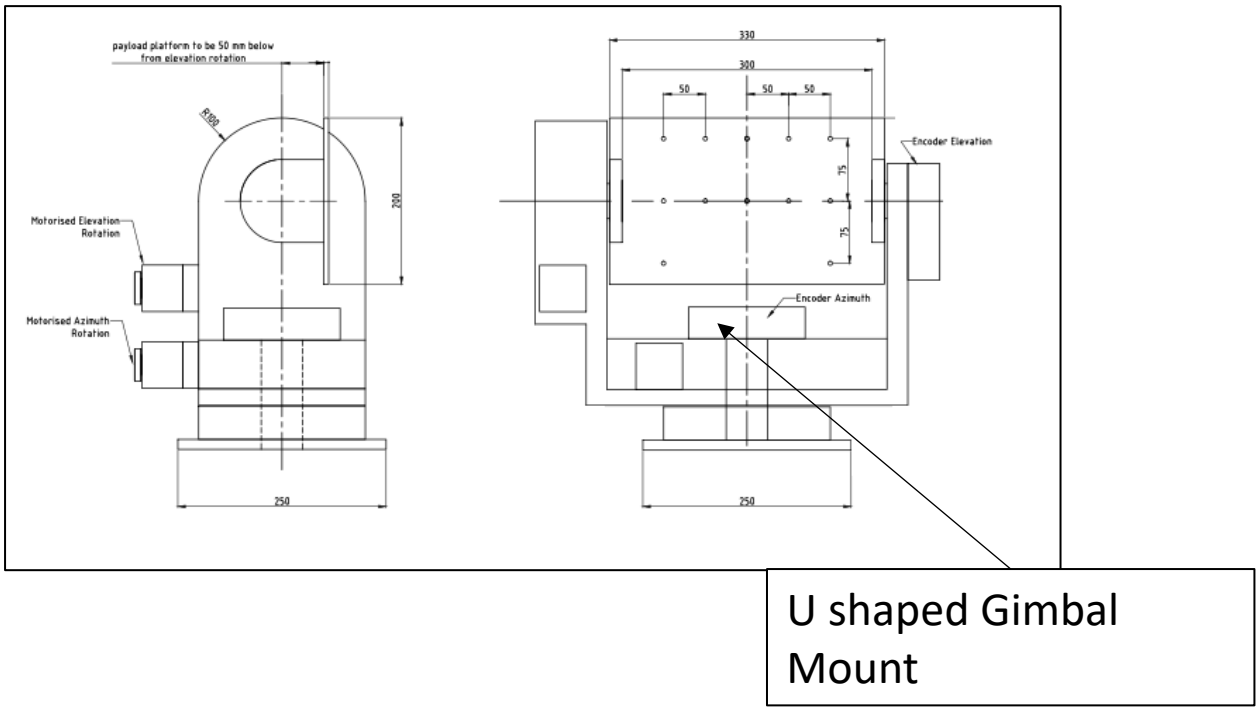


Figure 21 Two AXIS Turn Table

- Operation: Encoder based motorized rotation. Gear Mechanism to be avoided.
- Number of rotation axis: 2 nos.
- Azimuth (Az) Rotation Travel: 360 deg
- Elevation (El) Rotation Travel: 180 deg
- Azimuth& Elevation Rotation resolution: 0.5 arc second
- Azimuth& Elevation Rotation positioning accuracy: 1 arc second
- Orthogonality between Azimuth & Elevation rotation Axis – 1 arc second.
- Az & El Configuration: Elevation rotation must be built above Azimuth rotation. Refer the image below.
- Shape: U-Shaped Gimbal. Refer the following image "2-axis U shaped motorised gimbal mount reference". 2-axis gimbal mount shall be similar to the image.
- Payload capacity: 10 kg

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